

Alternating direction method of multipliers for polynomial optimization

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21st European Control Conference (ECC)
June 14, 2023

Outline

- 1 Introduction: polynomial optimization & ADMM
- 2 Problem transformation
- 3 ADMM4POP iterations
- 4 Convergence analysis
- 5 Numerical experiments
- 6 Conclusion

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Polynomial optimization

A **polynomial optimization problem (POP)** is a problem in which both objective function and constraints are multivariate polynomials.

$$\min_{x \in \mathbb{R}^n} f(x)$$

s.t.

$$g_i(x) \leq 0, \quad i = 1, \dots, m$$

$$h_i(x) = 0, \quad i = 1, \dots, m'$$

Polynomial optimization

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The diagram illustrates the components of a polynomial optimization problem (POP). It shows the objective function $f(x)$ and the constraints $g_i(x) \leq 0$ and $h_i(x) = 0$. All these expressions are circled in green. Green arrows point from the text "Polynomials!" to each of these circled expressions, indicating that they are all polynomials. The optimization problem is written as:

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & f(x) \\ \text{s.t.} & \\ & g_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, m' \end{array}$$

Polynomials!

POPs arise in many engineering applications:

- Set-membership estimation and identification (Cerone et al., 2014);
- Direct data-driven control (Cerone et al., 2017);
- Model predictive control (Canale et al., 2014), (Harinath et al., 2016);
- Localization in narrowband internet-of-things (Sedighi et al., 2021);
- Routing games (Zhu & Savla, 2022);
- Estimation of Lipschitz constants in deep neural networks (Chen et al., 2020);
- ... many others!

In general, POPs are non-convex and NP-hard.

Solutions proposed in the literature are:

- ① based on semidefinite programming (SDP) relaxations $\rightarrow$ get global optima, but computationally prohibitive for large problems;
- ② "general-purpose" local methods: interior-point method, sequential quadratic programming, $\dots \rightarrow$ no theoretical guarantee of convergence.

In general, POPs are non-convex and NP-hard.

In this work, we provide a memory and computationally efficient local algorithm with guaranteed convergence.

Alternating direction method of multipliers

Alternating Directions Method of Multipliers (ADMM) is a first-order algorithm. It applies to problems of the kind

$$\begin{aligned} \min_{x,z \in \mathbb{R}^n} \quad & F(x) + G(z) \\ \text{s.t.} \quad & x = z. \end{aligned}$$

Alternating direction method of multipliers

Defined the augmented Lagrangian

$$L_\rho(x, z) = F(x) + G(z) + \rho u^\top (x - z) + \frac{\rho}{2} \|x - z\|_2^2,$$

it consists in the iteration of:

- ① x-update: $x_{\tau+1} = \arg \min_{x \in \mathbb{R}^n} L_\rho(x, z) = \arg \min_{x \in \mathbb{R}^n} F(x) + \frac{\rho}{2} \|x - z_\tau + u_\tau\|_2^2,$
- ② z-update: $z_{\tau+1} = \arg \min_{z \in \mathbb{R}^n} L_\rho(x, z) = \arg \min_{z \in \mathbb{R}^n} G(z) + \frac{\rho}{2} \|x_{\tau+1} - z + u_\tau\|_2^2,$
- ③ u-update: $u_{\tau+1} = u_\tau + x_{\tau+1} - z_{\tau+1}.$

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Quadratic optimization problem

By suitably introducing **slack variables**, we can transform any POP to a quadratic optimization problem (QOP) of the kind

$$\begin{aligned} \min_{x \in \mathbb{R}^{n'}} \quad & \frac{1}{2} x^\top A x + a^\top x \\ \text{s.t.} \quad & Bx - b \leq 0 \\ & Cx - c = 0 \\ & x_i x_j = x_k, \quad (i, j, k) \in \mathcal{B} \end{aligned}$$

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$$\begin{aligned} \min_{x \in \mathbb{R}^{n'}} \quad & \boxed{\frac{1}{2}x^\top Ax + a^\top x} \quad \leftarrow \begin{array}{l} A \succeq 0 \\ \text{convex} \end{array} \\ \text{s.t.} \quad & Bx - b \leq 0 \\ & Cx - c = 0 \\ & x_i x_j = x_k, \quad (i, j, k) \in \mathcal{B} \end{aligned}$$

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Quadratic optimization problem

- The objective function is convex and quadratic: $A \succeq 0$.
Possibly linear, i.e., $A = 0$.
- Across different tuples in $\mathcal{B}$, entries indices i, j, k are **never repeated**.

Choosing

$$F(x) = \frac{1}{2}x^\top Ax + a^\top x, \quad \text{dom}(F) = \mathcal{D}$$

$$G(z) = \mathbb{1}_{\mathcal{P}}(z) = \{0 \text{ if } z \in \mathcal{P}, +\infty \text{ otherwise}\}$$

the problem is recast to a form suitable for application of ADMM.

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The x-update

x-update amounts for the solution of

$$\begin{aligned}x_{\tau+1} = \arg \min_{x \in \mathbb{R}^{n'}} & \frac{1}{2} x^\top (A + \rho I_{n'}) x + x^\top [a + \rho(u_\tau - z_\tau)] \\ \text{s.t.} & \\ & Bx - b \leq 0, \\ & Cx - c = 0\end{aligned}$$

which is a **convex quadratic program!**

Moreover, if no inequalities are present:

- the KKT conditions is a linear system of equations.
- the matrix $[A + \rho I, C^\top; C, 0]$ to be inverted can be factored before iterations starts.

The z-update

The z-update is the projection:

$$z_{\tau+1} = \arg \min_{z \in \mathcal{P}} \|z - (x_{\tau+1} + u_{\tau})\|_2^2.$$

This is non-convex and NP-hard in general. But **exploiting separability of $\mathcal{P}$** :

$$\begin{aligned} (z_{\tau+1})_t &= \arg \min_{z_t \in \mathbb{R}^3} (z_i - v_i)^2 + (z_j - v_j)^2 + (z_k - v_k)^2 \\ \text{s.t.} \quad & z_i z_j = z_k. \end{aligned}$$

with $v_t = (v_i, v_j, v_k) \doteq (x_{\tau+1} + u_{\tau})_t \in \mathbb{R}^3$, for all $t = (i, j, k) \in \mathcal{B}$.

The z-update

- The z-update problem has been decomposed into $|\mathcal{B}|$ smaller problems.
- Each of these can be solved in closed form by solving

$$z_j^5 + c_4 z_j^4 + c_3 z_j^3 + c_2 z_j^2 + c_1 z_j + c_0 = 0$$
$$z_i = \frac{v_i + z_j v_k}{1 + z_j^2}, \quad z_k = z_i z_j.$$

and taking the real solution obtain lower objective.

The solution is obtained in **closed form** after computing eigenvalues of a 5×5 matrix.

- The $|\mathcal{B}|$ small problems can be solved **in parallel**.

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Convergence result

We analyze a simplified case, in which: (i) no inequality constraints are present, and (ii) the equalities are accounted for in a penalized fashion, i.e.,

$$\min_{x \in \mathbb{R}^{n'}} \quad \frac{1}{2}x^\top Ax + a^\top x + \gamma \|Cx - c\|_2^2 + \mathbb{1}_P(x). \quad (\star)$$

We refer to the proposed algorithm applied on this simplified problem as "relaxed ADMM4POP".

Theorem (Convergence of relaxed ADMM4POP)

The sequence (x_k, z_k, u_k) generated by relaxed ADMM4POP converges to a point (x^, z^*, u^*) , and x^* is a stationary point of $(\star)$.*

The proof of convergence is based on a previous result from (Li & Pong, 2015).

According to their result, our result holds if the following conditions are met:

- A1. F is semi-algebraic, twice continuously differentiable on $\mathbb{R}^n$ with a bounded, positive semidefinite Hessian $\nabla^2 F$;
- A2. G is a semi-algebraic, proper, lower semicontinuous function;
- A3. there exists $0 < \zeta < \rho$ such that $F(x) - \frac{1}{2\zeta} \|\nabla F(x)\|^2$ is lower bounded;
- A4. the design parameter ρ is designed so that $\rho I_n - \sqrt{2} \nabla^2 F(x) \succeq 0$.

We show that all of them are verified.

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Example 1

We consider the following POP:

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x_1^2 x_2^2 + x_1^2 + q_1 x_1 + x_2^2 + x_2 x_3 + q_2 x_2 + x_3^2 + q_3 x_3 \\ \text{s.t.} \quad & x_2 x_3 + x_1 = 10 \end{aligned}$$

where $q_1 \in [4, 6]$, $q_2 \in [-8, -6]$, and $q_3 \in [1, 3]$ are generated uniformly at random.

	Error	Runtime (ms)		
	mean	mean	min	max
SDP	—	63.5	59.0	88.5
IPM	2.5×10^{-5}	12.3	7.7	21.8
ADMM4POP cns	6.5×10^{-5}	1.6	1.5	7.9
ADMM4POP rel	4.2×10^{-4}	1.7	1.6	8.9

Example 2

We consider the discrete-time dynamic system $y(k+1) = \alpha y(k) + \beta u(k+1)$, and we aim at recovering the parameters α and β given: (i) $u \in \mathbb{R}^K$, (ii) noisy measurements of the output $w \doteq y + \eta$. We solve:

$$\begin{aligned} \min_{\alpha, \beta \in \mathbb{R}, y \in \mathbb{R}^K} \quad & \|y - w\|_2^2 \\ \text{s.t.} \quad & \\ & y(k+1) = \alpha y(k) + \beta u(k+1) \quad k = 1, \dots, K-1. \end{aligned}$$

Example 2

	Error Mean	Runtime (s)		
		mean	min	max
SDP	6.1×10^{-4}	10.16	5.87	22.99
IPM	3.2×10^{-4}	3.97	1.34	36.73
ADMM4POP cns	3.6×10^{-4}	2.53	0.76	21.45
ADMM4POP rel	3.9×10^{-4}	0.62	0.40	1.61

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Conclusions

- We provide a reformulation of any POP as a QOP with special convex + bilinear structure.
- We propose an ADMM-based algorithm to find a solution to the reformulated problem.
- Along with the main algorithm, we propose a simplified version (relaxed ADMM4POP), which is shown to always converge and is computationally less expensive.
- We show, through numerical examples, that the proposed algorithm is more computationally efficient and less memory expensive than current state-of-the-art algorithms.